

CHAPTER 2

PLANNING / DECISION OF CAPITAL STRUCTURE

1. Meaning

Planning of Capital Structure means deciding the proper proportion of debt, preference capital and equity in the total long-term finance of a firm.

Main Objective

To determine an optimum debt–equity mix which:

- Maximises EPS and shareholders' wealth
- Minimises overall cost of capital
- Maintains reasonable financial risk
- Maximises the value of the firm

OPTIMUM CAPITAL STRUCTURE = Best balance of Return + Cost + Risk + Firm Value

2. Financial Decision Chart

FINANCIAL MANAGEMENT



INVESTMENT
DECISION

FINANCING
DECISION

DIVIDEND
DECISION



CAPITAL STRUCTURE DECISION



DEBT–EQUITY MIX



Effect on EPS

Effect on Risk



COST OF CAPITAL



VALUE OF FIRM

BASIC LOGIC: Debt–Equity Mix → EPS & Risk → Cost of Capital → Value of Firm

TRADING ON EQUITY

MEANING, DEFINITIONS & TYPES

3. Meaning of Trading on Equity

Trading on Equity means using fixed-cost funds such as debt and preference capital along with equity with the objective of increasing the return to equity shareholders.

It is also closely associated with financial leverage.

TRADING ON EQUITY = Use of Fixed-Cost Funds to Improve Equity Shareholders' Return

Definitions

Gerstenberg: Trading on equity occurs when a business uses borrowed capital together with owned capital in carrying on its business.

Guthmann and Dougall: The use of borrowed funds or preference capital carrying fixed financial charges for financing is known as trading on equity.

4. Types of Trading on Equity

A. TRADING ON THIN EQUITY

- Debt / fixed-charge capital is high and equity capital is low.
- Debt > Equity
- High financial leverage
- Higher potential EPS
- Higher financial risk

B. TRADING ON THICK EQUITY

- Equity capital is high and debt is low.
- Equity > Debt
- Low financial leverage
- Lower financial risk
- Lower possibility of magnifying EPS through debt

THIN EQUITY → Heavier Dependence on Borrowed Funds



THICK EQUITY → Heavier Dependence on Owners' Funds

TRADING ON EQUITY

OBJECTIVES & IMPORTANCE

5. Objectives of Trading on Equity

1. Increase EPS
2. Increase return to equity shareholders
3. Take advantage of low-cost debt
4. Avoid unnecessary issue of additional equity shares
5. Maintain shareholders' control
6. Obtain tax advantage of interest on debt

Basic Purpose: Use fixed-cost financing when the return generated by the business is sufficient to produce an additional benefit for equity shareholders.

6. Importance of Trading on Equity

1. **Higher EPS** — Successful use of debt may increase earnings available per equity share.
2. **Higher Return on Equity** — Equity shareholders can earn a higher return when the firm earns more on borrowed funds than their fixed cost.
3. **Control** — Debt financing does not normally dilute equity ownership.
4. **Tax Benefit** — Interest on debt is generally deductible before calculating taxable profit.
5. **Capital Structure Planning** — It helps management decide the appropriate debt–equity combination.

Important Limitation: More debt may increase EPS, but it also increases financial risk and fixed interest obligations.

Concept Flow

Use Debt / Preference Capital with Equity



Fixed Financial Charges Arise



Potential Increase in EPS / ROE



But Financial Risk Also Increases

TRADING ON EQUITY MEASUREMENT & EFFECTS

7. Measurement of Trading on Equity

Trading on Equity can be studied through the following measures:

A. Debt–Equity Ratio = Debt / Equity

Higher ratio → Greater use of financial leverage.

B. Degree of Financial Leverage (DFL) = EBIT / EBT

or $DFL = EBIT / (EBIT - \text{Interest})$

If higher DFL means EPS is more sensitive to changes in EBIT.

C. EPS Comparison: All Equity Plan vs. Equity + Debt Plan

If debt increases EPS → Trading on Equity is favourable.

8. Effects of Trading on Equity

Trading on Equity magnifies the effect of changes in EBIT on EPS; greater debt generally means greater sensitivity of EPS and greater financial risk.

FAVOURABLE	NEUTRAL	UNFAVOURABLE
ROI > Interest Rate Debt financing increases EPS Positive Financial Leverage	ROI = Interest Rate No EPS advantage from debt	ROI < Interest Rate Debt financing reduces EPS Negative Financial Leverage

ROI > Interest → Favourable → Higher Debt may raise EPS

ROI = Interest → Neutral → No EPS advantage

ROI < Interest → Unfavourable → Higher Debt may lower EPS

PRACTICAL PROBLEM

FAVOURABLE TRADING ON EQUITY

9. Given Information

A company requires ₹10,00,000 and is comparing two financing plans:

PLAN A – ALL EQUITY

- Equity = ₹10,00,000

PLAN B – EQUITY + DEBT

- Equity = ₹5,00,000
- Debt = ₹5,00,000 @ 10%

Share Price = ₹10 | Tax Rate = 30% | Situation I: EBIT = ₹1,50,000

Preliminary Calculations

Particulars	Plan A	Plan B
Number of Equity Shares	$₹10,00,000 \div ₹10 = 1,00,000$	$₹5,00,000 \div ₹10 = 50,000$
Interest	Nil	$₹5,00,000 \times 10\% = ₹50,000$

Situation I – EBIT = ₹1,50,000

Particulars	All Equity	Equity + Debt
EBIT	₹1,50,000	₹1,50,000
Less: Interest	Nil	₹50,000
EBT	₹1,50,000	₹1,00,000
Tax / EAT	$₹1,50,000 \times 70\% = ₹1,05,000$	$₹1,00,000 \times 70\% = ₹70,000$
No. of Shares	1,00,000	50,000
EPS	₹1.05	₹1.40

Debt Plan EPS ₹1.40 > All Equity EPS ₹1.05

$ROI = ₹1,50,000 / ₹10,00,000 \times 100 = 15\%$

ROI 15% > Interest Rate 10%



TRADING ON EQUITY IS FAVOURABLE

PRACTICAL PROBLEM

UNFAVOURABLE TRADING ON EQUITY

10. Situation II – EBIT Falls to ₹80,000

The financing plans, share price and tax rate remain the same. Only EBIT falls to ₹80,000.

Particulars	All Equity Plan	Equity + Debt Plan
EBIT	₹80,000	₹80,000
Less: Interest	Nil	₹50,000
EBT	₹80,000	₹30,000
EAT @ 70%	₹56,000	₹21,000
No. of Shares	1,00,000	50,000
EPS	₹0.56	₹0.42

Debt Plan EPS ₹0.42 < All Equity EPS ₹0.56

ROI Test

$$\text{ROI} = ₹80,000 / ₹10,00,000 \times 100 = 8\%$$

ROI 8% < Interest Rate 10%



TRADING ON EQUITY IS UNFAVOURABLE

Why does EPS fall under Debt?

EBIT Falls



Interest ₹50,000 remains Fixed



Lower EBT and EAT for Equity Shareholders



Debt Plan EPS becomes lower than All-Equity EPS

CHAPTER 2

QUICK REVISION & DECISION LOGIC

	Quick Revision
Capital Structure Decision	Decision regarding Debt–Equity Mix
Trading on Equity	Use fixed-cost finance to improve equity shareholders' return
Thin Equity	More Debt + Less Equity → High Risk
Thick Equity	More Equity + Less Debt → Low Risk

Favourable / Neutral / Unfavourable Rule

ROI > Interest Rate



Favourable Trading on Equity → Debt may increase EPS

ROI = Interest Rate



Neutral Trading on Equity → No EPS advantage

ROI < Interest Rate



Unfavourable Trading on Equity → Debt may reduce EPS

Complete Decision Logic

Choose Debt–Equity Mix



Measure EPS Benefit + Financial Risk



Compare ROI with Interest Cost



Assess Cost of Capital



Select Capital Structure that Supports Firm Value

GOLDEN RULE: Debt can increase EPS, but debt also increases financial risk.